

Quantum Information Theory

Peter Zeman

NMMB537

Preface

These are lecture notes for the course NMMB537 Quantum Information Theory, taught at Faculty of Mathematics and Physics, Charles University. The lecture notes are mostly inspired by the course NMAK14020U Quantum Information Theory, offered at University of Copenhagen.

Contents

Preface	i
1 Quantum States and Measurements	1
1.1 Finite-dimensional Hilbert Spaces	1
1.2 Quantum States	3
1.3 Bloch Sphere	4
1.4 Measurements	6
1.5 Bloch Sphere Revisited	8
1.6 Observables	8
2 Multiple Quantum Systems	11
2.1 Multiple Systems	11
2.2 The Partial Trace	14
2.3 Purification	18
2.4 Schmidt Decomposition	20
2.5 Entanglement	22

Chapter 1

Quantum States and Measurements

We start by describing a precise mathematical framework for quantum states.

1.1 Finite-dimensional Hilbert Spaces

Axiom 1 (Hilbert space). To every quantum system we associate a Hilbert space $\mathcal{H}$.

We only consider finite-dimensional Hilbert spaces in this course. If $\dim(\mathcal{H}) = d$, then $\mathcal{H}$ is isomorphic to

$$\mathbb{C}^d = \left\{ \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_d \end{pmatrix} : \psi_i \in \mathbb{C} \right\}$$

with the standard inner product

$$\langle \psi | \phi \rangle = \sum_{i=1}^d \overline{\psi_i} \phi_i.$$

Bra-ket notation. We write $|\psi\rangle \in \mathcal{H}$ for any vector. By $\langle\psi| \in \mathcal{H}^\dagger$, we mean the linear map $\mathcal{H} \rightarrow \mathbb{C}$ defined by $\langle\psi|(|\phi\rangle) = \langle\psi | \phi\rangle$. In other words, $\langle\psi|$ is the row vector corresponding to the conjugate transpose of $|\psi\rangle$. We denote the standard basis of $\mathbb{C}^d$ by $|0\rangle, |1\rangle, \dots, |d-1\rangle$. With coordinates numbered $1, \dots, d$, the vector $|i\rangle$ has 1 in coordinate $i+1$ and 0 elsewhere.

Linear algebra notation. Every basis of a Hilbert space that we consider will be orthonormal. If there is no confusion, we will use the terms linear map and matrix interchangeably.

By $\text{Lin}(\mathcal{H}, \mathcal{K})$, we denote the set of linear maps $\mathcal{H} \rightarrow \mathcal{K}$. When we fix the standard bases, we can express $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$ as

$$M = \sum_{i,j} M_{ij} |i\rangle\langle j|.$$

Note that $\langle i|M|j\rangle = M_{ij} \in \mathbb{C}$.

Exercise 1.1. Write $\sum_{i,j \in \{0,1\}} ij|i\rangle\langle j|$ as a matrix.

If $M \in \text{Lin}(\mathcal{H}, \mathcal{H}) = \text{Lin}(\mathcal{H})$, then the trace with respect to the standard basis is

$$\text{tr}[M] = \sum_i M_{ii} = \sum_i \langle i|M|i\rangle.$$

An important fact about trace is that it is in fact independent of the basis. Recall also that $\text{tr}[MN] = \text{tr}[NM]$.

Here are some important types of matrices:

- $M \in \text{Lin}(\mathcal{H})$ is *Hermitian* if $M^\dagger = M$.
- $P \in \text{Lin}(\mathcal{H})$ is *positive* if $\langle \psi | P | \psi \rangle \geq 0$, for all $|\psi\rangle \in \mathcal{H}$. $\text{PSD}(\mathcal{H})$ is the class of all positive matrices. We also use the notation $P \geq 0$ to indicate that P is positive. If $M, N \in \text{Lin}(\mathcal{H})$, then we define $M \geq N$ if $M - N \geq 0$, that is, if $M - N$ is positive.
- $U \in \text{Lin}(\mathcal{H})$ is *unitary* if $U^\dagger U = U U^\dagger = I$. $\mathcal{U}(\mathcal{H})$ is the class of all unitary matrices.
- $V \in \text{Lin}(\mathcal{H}, \mathcal{K})$ is an *isometry* if $V^\dagger V = I$.
- $P \in \text{Lin}(\mathcal{H})$ is an *orthogonal projection* if $P^\dagger = P$ and $P^2 = P$.

Exercise 1.2 (Why isometries preserve geometry). Show that $V \in \text{Lin}(\mathcal{H}, \mathcal{K})$ satisfies $V^\dagger V = I$ if and only if it preserves inner products. Deduce that it preserves lengths and distances.

Exercise 1.3 (Why these operators are orthogonal projections). Show that the conditions $P^2 = P$ and $P^\dagger = P$ mean that P is the orthogonal projection onto its image $W = \text{ran}(P)$.

Exercise 1.4. Are the following matrices Hermitian, PSD, unitary, or orthogonal projections?

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} e^{i\pi/4} & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix}.$$

Theorem 1.5 (Spectral theorem for Hermitian matrices). *If $M \in \text{Lin}(\mathcal{H})$ is Hermitian $d \times d$ matrix, then M has eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$ and there is an orthonormal basis of eigenvectors $\{|\psi_i\rangle\}_{i=1}^d$ such that*

$$M = \sum_{i=1}^d \lambda_i |\psi_i\rangle\langle\psi_i|.$$

Recall that the eigenvalues can have multiple occurrences and the basis of eigenvectors may not be unique. Indeed, for instance,

$$I = \sum_{i=1}^d |\psi_i\rangle\langle\psi_i|$$

is true for any orthonormal basis $\{|\psi_i\rangle\}_{i=1}^d$.

This allows us to define functions of Hermitian matrices. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function, then we can define

$$f(M) = \sum_{i=1}^d f(\lambda_i) |\psi_i\rangle\langle\psi_i|.$$

For instance, if $M \geq 0$, the same construction with $f(t) = \sqrt{t}$ on $[0, \infty)$ defines its positive square root:

$$\sqrt{M} = \sum_{i=1}^d \sqrt{\lambda_i} |\psi_i\rangle\langle\psi_i|.$$

Note that if P is an orthogonal projection, then the condition $P^2 = P$ implies that the eigenvalues are in $\{0, 1\}$. Thus, we can write $P = \sum_{i=1}^r |\psi_i\rangle\langle\psi_i|$. Here, $r = \text{rank}(P)$ of the orthogonal projection P .

Theorem 1.6 (Characterization of positive matrices). *If $P \in \text{Lin}(\mathcal{H})$, then the following are equivalent:*

- (a) P is positive, i.e., $\langle\psi|P|\psi\rangle \geq 0$, for all $|\psi\rangle \in \mathcal{H}$.
- (b) $P^\dagger = P$ and all eigenvalues are nonnegative.
- (c) There is $M \in \text{Lin}(\mathcal{H})$ such that $P = M^\dagger M$.
- (d) $\text{tr}[PQ] \geq 0$, for every $Q \in \text{PSD}(\mathcal{H})$.

Exercise 1.7. Prove that if $P \in \text{PSD}(\mathcal{H})$ and $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$, then $MPM^\dagger$ is positive.

1.2 Quantum States

Definition 1.8. A *density matrix* is a positive operator $\rho \in \text{PSD}(\mathcal{H})$ with $\text{tr}[\rho] = 1$. We put $\mathcal{S}(\mathcal{H}) = \{\rho \in \text{PSD}(\mathcal{H}) : \text{tr}[\rho] = 1\}$.

Axiom 2. The state of a quantum system with Hilbert space $\mathcal{H}$ is described by a density matrix ρ . We will refer to ρ as a quantum state.

Example 1.9 (classical states). In probability theory we have a finite set of outcomes Σ with a probability distribution p such that $p(x) \geq 0$, for $x \in \Sigma$, and $\sum_{x \in \Sigma} p(x) = 1$. If $\mathcal{H}$ has an orthonormal basis $\{|x\rangle\}_{x \in \Sigma}$, then

$$\rho = \sum_{x \in \Sigma} p(x) |x\rangle\langle x|$$

is a density matrix. Conversely, any diagonal density matrix corresponds to a probability distribution.

Example 1.10 (pure states). Let $|\psi\rangle \in \mathcal{H}$ be a unit vector. Then $\rho = |\psi\rangle\langle\psi|$ is a density matrix since $\text{tr}[|\psi\rangle\langle\psi|] = \text{tr}[\langle\psi|\psi\rangle] = 1$. If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$|\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\bar{\beta} \\ \bar{\alpha}\beta & |\beta|^2 \end{pmatrix}.$$

Note that replacing $|\psi\rangle$ by $e^{i\theta}|\psi\rangle$ does not change the density matrix.

Exercise 1.11. Is $\rho = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ a pure state?

Solution. Yes, since $\rho = |+\rangle\langle+|$, where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. □

Example 1.12 (general state). Let ρ be a general quantum state. By Theorem 1.6, we can write

$$\rho = \sum_{i=1}^d p_i |\psi_i\rangle\langle\psi_i|,$$

where $p_i \geq 0$. Since ρ is diagonal in the basis $\{|\psi_i\rangle\}_{i=1}^d$ and trace is independent of the chosen basis, $\text{tr}[\rho] = 1$ implies that $\sum_{i=1}^d p_i = 1$. So, the eigenvalues p_i 's define a probability distribution. We may interpret ρ as if we have the pure state $|\psi_i\rangle\langle\psi_i|$ with probability p_i . We call ρ a *mixed state* if it is not pure.

If the spectrum is nondegenerate, this spectral decomposition is unique up to ordering and phases of the eigenvectors. Nevertheless, a mixed state admits other *ensemble decompositions* into pure states that need not be orthogonal, with different probabilities and possibly different numbers of states.

Exercise 1.13. What is the state corresponding to 50% chance of $|0\rangle$ and 50% chance of $|+\rangle$?

Solution.

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}.$$

Note that the eigenvalues of ρ are $1/2 + \sqrt{2}/4$ and $1/2 - \sqrt{2}/4$, so the spectral decomposition of ρ gives a completely different decomposition of ρ compared to the one we started with. $\square$

Definition 1.14. The state $\tau = \frac{1}{d}I$ is called the *maximally mixed state*.

Exercise 1.15 (Different preparations of the same state). Compare the state obtained by choosing $|0\rangle$ or $|1\rangle$ with equal probabilities with the state obtained by choosing $|+\rangle$ or $|-\rangle$ with equal probabilities, where $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Solution. The first preparation gives

$$\rho_Z = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

For the second preparation, the off-diagonal entries cancel:

$$\rho_X = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{I}{2}.$$

Thus both preparations give the same maximally mixed state. In contrast, preparing the superposition $|+\rangle$ every time gives $|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Adding state vectors to form a superposition and averaging density matrices to describe a random preparation are different operations. $\square$

1.3 Bloch Sphere

How does the set $\mathcal{S}(\mathcal{H})$ look like? For two states $\rho_1, \rho_2 \in \mathcal{S}(\mathcal{H})$ and $t \in [0, 1]$, the convex combination $t\rho_1 + (1-t)\rho_2$ is also a state. In fact, we have the following lemma.

Lemma 1.16. *The set $\mathcal{S}(\mathcal{H}) \subseteq \text{Lin}(\mathcal{H})$ is convex. Moreover, the extreme points of $\mathcal{S}(\mathcal{H})$ are exactly the pure states.*

For the particular case of one qubit, there is a visualization known as the Bloch sphere. Hermitian 2×2 matrices have the following (real) basis:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The last three matrices are known as Pauli matrices. Here are some of their properties:

$$X^2 = Y^2 = Z^2 = I, \quad XY = iZ = -YX, \quad \text{tr}[X] = \text{tr}[Y] = \text{tr}[Z] = 0.$$

In particular, Pauli matrices are unitary and have eigenvalues ± 1 .

Every 2×2 Hermitian matrix can be written in a unique way as a linear combination of the matrices I, X, Y, Z with all coefficients being real numbers. An arbitrary 2×2 Hermitian matrix ρ with $\text{tr}[\rho] = 1$ can be expressed as

$$\rho = \frac{1}{2}(I + xX + yY + zZ) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix}. \quad (1.1)$$

The coefficient $1/2$ in front of the I is fixed by the condition $\text{tr}[\rho] = 1$. Further, $\text{tr}[\rho] = 1$ implies that the eigenvalues of ρ are λ and $1 - \lambda$, for some $\lambda \in \mathbb{R}$. If we want ρ to be a quantum state, it also has to be positive, which is equivalent to requiring that $\lambda \geq 0$ and $1 - \lambda \geq 0$. Now, λ and $1 - \lambda$ are nonnegative if and only if $\det(\rho) = \lambda(1 - \lambda) \geq 0$. We compute

$$\det(\rho) = \frac{1}{4}((1+z)(1-z) - (x+iy)(x-iy)) = \frac{1}{4}(1 - x^2 - y^2 - z^2).$$

If we let $r = (x, y, z) \in \mathbb{R}^3$, then $\rho \geq 0$ if and only if $\|r\| \leq 1$. If ρ is a pure state, then $\text{rank}(\rho) = 1$ and $\det(\rho) = 0$, which is equivalent to $\|r\| = 1$. Thus, pure states correspond exactly to the unit sphere in $\mathbb{R}^3$ and mixed states correspond to the open unit ball.

Exercise 1.17. What is the pure state $|\psi\rangle$ and its density matrix $|\psi\rangle\langle\psi|$ corresponding to $r_1 = (0, 1, 0)$ and to $r_2 = (0, -1, 0)$.

Solution. By plugging r_1 into Eq. (1.1), we get

$$\rho_1 = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}.$$

We can easily calculate, that the eigenvector corresponding to the eigenvalue 1 of ρ_1 is $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$. By Theorem 1.5, we get $\rho_1 = |\psi_1\rangle\langle\psi_1|$. Similarly, we get that $\rho_2 = |\psi_2\rangle\langle\psi_2|$, where $|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$. $\square$

We have the following important bases of pure states that correspond to the intersection points of x, y, z -axes with the Bloch sphere:

- Z -basis: $\{|0\rangle, |1\rangle\}$, $r = (0, 0, 1)$, $-r = (0, 0, -1)$.
- X -basis: $\{|+\rangle, |-\rangle\}$, $r = (1, 0, 0)$, $-r = (-1, 0, 0)$
- Y -basis: $\{|+i\rangle, |-i\rangle\}$, $r = (0, 1, 0)$, $-r = (0, -1, 0)$.

Exercise 1.18 (Reading a Bloch vector). Find the Bloch vector of the state $\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}$ from Exercise 1.13. Is it pure or mixed?

Solution. Comparing entries with Eq. (1.1), we obtain

$$\frac{1+z}{2} = \frac{3}{4}, \quad \frac{x-iy}{2} = \frac{1}{4}, \quad \text{so} \quad r = \left(\frac{1}{2}, 0, \frac{1}{2}\right).$$

Its length is $\|r\| = 1/\sqrt{2} < 1$, so the state is mixed. The vector is also the average of the Bloch vectors of $|0\rangle$ and $|+\rangle$:

$$r = \frac{1}{2}(0, 0, 1) + \frac{1}{2}(1, 0, 0).$$

This illustrates how a probabilistic mixture becomes a convex combination of Bloch vectors. $\square$

1.4 Measurements

Definition 1.19. A *measurement* or a *positive operator valued measure (POVM)* on a Hilbert space $\mathcal{H}$ with a finite set of outcomes Ω is a function $\mu : \Omega \rightarrow \text{PSD}(\mathcal{H})$ such that $\sum_{x \in \Omega} \mu(x) = I$. If $\mu(x)$ are orthogonal projections, then the measurement is called *projective*.

Axiom 3. If we measure a state ρ using μ , then the probability of getting an outcome $x \in \Omega$ is $p(x) = \text{tr}[\mu(x)\rho]$.

Remark 1.20. In general in quantum mechanics, it is not possible to perform measurements on a state without changing it. However, for now, we will not consider what happens with a state after the measurement (we can think that it is destroyed). We only get classical information about which outcome occurred. We will discuss post-measurement states later in the course.

Exercise 1.21. Prove that if ρ is a quantum state and μ is a measurement, then $p(x) = \text{tr}[\mu(x)\rho]$ defines a probability distribution on Ω .

Solution. By Theorem 1.6(d), we have $\text{tr}[\mu(x)\rho] \geq 0$ since both $\mu(x)$ and ρ are positive. By the linearity of trace,

$$\sum_{x \in \Omega} \text{tr}[\mu(x)\rho] = \text{tr} \left[\sum_{x \in \Omega} \mu(x)\rho \right] = \text{tr}[I\rho] = 1.$$

$\square$

Example 1.22 (basis measurement). Let $\{|\psi_i\rangle\}_{i=1}^d$ be an orthonormal basis and let $\mu(i) = |\psi_i\rangle\langle\psi_i|$ be the orthogonal projection onto $|\psi_i\rangle$. We get

$$p(i) = \text{tr}[|\psi_i\rangle\langle\psi_i|\rho] = \text{tr}[\langle\psi_i|\rho|\psi_i\rangle] = \langle\psi_i|\rho|\psi_i\rangle.$$

In particular, if $\rho = |\phi\rangle\langle\phi|$ is a pure state, then we get

$$p(i) = \langle\psi_i | \phi\rangle\langle\phi | \psi_i\rangle = |\langle\psi_i | \phi\rangle|^2.$$

Exercise 1.23. What are the outcome probabilities when we measure $|0\rangle$ the X -basis?

Solution. We get $|+\rangle$ with probability $|\langle + | 0 \rangle|^2 = 1/2$ and $|-\rangle$ with probability $|\langle - | 0 \rangle|^2 = 1/2$. $\square$

Exercise 1.24 (Comparing two states by measurement). Let $\rho = |+\rangle\langle +|$ and $\tau = I/2$. Calculate the outcome probabilities for each state when measuring in the Z -basis and in the X -basis.

Solution. For a Z -basis measurement, the probabilities are the diagonal entries in the standard basis. Both states have diagonal entries $1/2, 1/2$. For the X -basis measurement, the probability of outcome $+$ is

$$\langle + | \rho | + \rangle = |\langle + | + \rangle|^2 = 1, \quad \langle + | \tau | + \rangle = \frac{1}{2} \langle + | + \rangle = \frac{1}{2}.$$

The probability of outcome $-$ is the complement. Altogether,

state	$p_Z(0)$	$p_Z(1)$	$p_X(+)$	$p_X(-)$
ρ	$1/2$	$1/2$	1	0
τ	$1/2$	$1/2$	$1/2$	$1/2$

The two states give the same statistics in the Z -basis, but different statistics in the X -basis. Knowing the probabilities for just one measurement need not determine the state. $\square$

Exercise 1.25 (A nonprojective measurement). Show that

$$\mu(0) = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \quad \mu(1) = \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix}$$

define a POVM that is not projective. What are the outcome probabilities on the state $|0\rangle$?

Solution. Both matrices are positive because they are diagonal with nonnegative entries, and $\mu(0) + \mu(1) = I$. They therefore define a POVM. However,

$$\mu(0)^2 = \begin{pmatrix} 9/16 & 0 \\ 0 & 1/16 \end{pmatrix} \neq \mu(0),$$

so the measurement is not projective. The Born rule gives

$$p(0) = \text{tr}[\mu(0)|0\rangle\langle 0|] = \langle 0 | \mu(0) | 0 \rangle = \frac{3}{4}, \quad p(1) = \frac{1}{4}.$$

One way to obtain these measurement statistics is to measure in the Z -basis and then flip the reported outcome with probability $1/4$. $\square$

1.5 Bloch Sphere Revisited

Given $r = (x, y, z)$ with $\|r\| = 1$, the positive matrices

$$\mu(0) := \rho(r) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} \quad \text{and} \quad \mu(1) := \rho(-r) = \frac{1}{2} \begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix}$$

define a general qubit basis measurement. Performing this measurement on a quantum state with Bloch vector $s = (x', y', z')$, then the probability of obtaining the outcome 0 is

$$p(0) = \frac{1}{2} + \frac{1}{2}(r \cdot s) = \frac{1}{2} + \frac{1}{2}(xx' + yy' + zz').$$

Geometrically $r \cdot s$ is the projection of the Bloch vector s of a quantum state onto the axis defining the measurement.

1.6 Observables

Sometimes it is convenient to reformulate measurements in terms of observables. A projective measurement is a measurement where all measurement operators are orthogonal projections, that is, we have a set of outcomes Ω and orthogonal projections $\{P_x : x \in \Omega\}$ such that $\sum_{x \in \Omega} P_x = I$. Suppose that $\Omega \subseteq \mathbb{R}$. The *observable* associated to the projective measurement is an operator $O \in \text{Lin}(\mathcal{H})$ given by

$$O = \sum_{x \in \Omega} x P_x.$$

Since x are real, O is Hermitian, and, conversely, each Hermitian operator O has a spectral decomposition of this form, thereby defining a projective measurement.

Remark 1.26. In physics, observables are often the preferred way to reason about measurements. Measuring the observable O is the same as performing the projective measurement defined by the spectral decomposition of O .

Example 1.27. Since $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, the observable Z corresponds to measuring in the Z -basis with outcomes $\{\pm 1\}$. Similarly, since $X = |+\rangle\langle +| - |-\rangle\langle -|$, measuring X corresponds to measuring in the X -basis with outcomes $\{\pm 1\}$. More generally, we may measure a qubit in the basis given by the antipodal points $r = (x, y, z)$ and $-r$ on the Bloch sphere with outcomes $\{\pm 1\}$. This gives an observable

$$\begin{aligned} O(r) &= \rho(r) - \rho(-r) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix} \\ &= \begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}. \end{aligned}$$

States and expectation values. In classical probability, a probability distribution p on a finite set Σ assigns an expectation to each real-valued random variable $a : \Sigma \rightarrow \mathbb{R}$:

$$\mathbb{E}_p[a] = \sum_{s \in \Sigma} p(s)a(s).$$

A quantum state plays a similar role: it assigns an expectation value to each observable. The observable specifies the measurement and its numerical outcomes, while the state determines their probabilities.

Let $O = \sum_{x \in \Omega} x P_x$ and let the state be ρ . The measurement outcome is an ordinary real-valued random variable, with $p_\rho(x) = \text{tr}[P_x \rho]$ by the Born rule. We write $\mathbb{E}_\rho[O]$ for its expectation. Taking the weighted average of the outcomes and using linearity of the trace gives

$$\begin{aligned} \mathbb{E}_\rho[O] &= \sum_{x \in \Omega} x p_\rho(x) = \sum_{x \in \Omega} x \text{tr}[P_x \rho] \\ &= \text{tr} \left[\left(\sum_{x \in \Omega} x P_x \right) \rho \right] = \text{tr}[O \rho] = \text{tr}[\rho O]. \end{aligned}$$

For a pure state $\rho = |\psi\rangle\langle\psi|$, this becomes

$$\mathbb{E}_\rho[O] = \text{tr}[O |\psi\rangle\langle\psi|] = \langle\psi|O|\psi\rangle.$$

Thus, we can view a state as an *expectation functional* $O \mapsto \mathbb{E}_\rho[O]$. Just like classical expectation, it is linear, normalized, and positive: for observables A, B and $a, b \in \mathbb{R}$,

$$\begin{aligned} \mathbb{E}_\rho[aA + bB] &= a\mathbb{E}_\rho[A] + b\mathbb{E}_\rho[B], \\ \mathbb{E}_\rho[I] &= 1, \\ A \geq 0 &\implies \mathbb{E}_\rho[A] \geq 0. \end{aligned}$$

The first two properties follow from linearity of trace and $\text{tr}[\rho] = 1$; the last follows from Theorem 1.6(d). Each expectation concerns the corresponding measurement on identically prepared systems.

Probabilities as expectations. An orthogonal projection P is itself an observable with possible outcomes in $\{0, 1\}$. It describes a yes-or-no question, and its expectation is the probability of the answer 1. In this sense, orthogonal projections play the role of indicator random variables. In particular, the probability of outcome x when measuring O is

$$p_\rho(x) = \text{tr}[\rho P_x] = \mathbb{E}_\rho[P_x].$$

The expectation functional therefore determines the entire outcome distribution of every projective measurement. More generally, the probability of outcome x for any POVM μ is $\mathbb{E}_\rho[\mu(x)]$.

Random preparations also obey the usual law of total expectation. If state ρ_i is prepared with probability p_i , then the average state is $\rho = \sum_i p_i \rho_i$ and

$$\mathbb{E}_\rho[O] = \text{tr} \left[\left(\sum_i p_i \rho_i \right) O \right] = \sum_i p_i \mathbb{E}_{\rho_i}[O].$$

For $O = \mu(x)$, the same identity is precisely the law of total probability for the measurement outcome x .

Exercise 1.28 (An expectation value is an average). For $\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}$ from Exercise 1.13, calculate the expectation value of Z using both the outcome probabilities and the trace formula.

Solution. The outcomes $+1$ and -1 correspond to $|0\rangle$ and $|1\rangle$, respectively. Their probabilities are $3/4$ and $1/4$, so

$$\mathbb{E}(\text{outcome}) = (+1)\frac{3}{4} + (-1)\frac{1}{4} = \frac{1}{2}.$$

The trace formula gives the same result:

$$\text{tr}[Z\rho] = \text{tr} \begin{pmatrix} 3/4 & 1/4 \\ -1/4 & -1/4 \end{pmatrix} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

A single measurement returns $+1$ or -1 . The expectation value $1/2$ describes the average over many repetitions on identically prepared systems. $\square$

Chapter 2

Multiple Quantum Systems

2.1 Multiple Systems

If classical random variables have outcome sets $\Sigma_1, \dots, \Sigma_n$, their joint distribution is a probability distribution on the product set

$$\Sigma_1 \times \dots \times \Sigma_n = \{(x_1, \dots, x_n) : x_j \in \Sigma_j\}.$$

In particular, the possible values of n bits are the tuples in $\{0, 1\}^n$.

The tensor product plays the corresponding role for quantum systems. We first define it for two finite-dimensional Hilbert spaces.

Tensor product of Hilbert spaces. For Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$, start with finite formal linear combinations of symbols $|\psi\rangle \otimes |\phi\rangle$, where $|\psi\rangle \in \mathcal{H}$ and $|\phi\rangle \in \mathcal{K}$. We identify such sums using the following *bilinearity relations*, for $\alpha, \beta \in \mathbb{C}$:

$$\begin{aligned} (\alpha|\psi_1\rangle + \beta|\psi_2\rangle) \otimes |\phi\rangle &= \alpha(|\psi_1\rangle \otimes |\phi\rangle) + \beta(|\psi_2\rangle \otimes |\phi\rangle), \\ |\psi\rangle \otimes (\alpha|\phi_1\rangle + \beta|\phi_2\rangle) &= \alpha(|\psi\rangle \otimes |\phi_1\rangle) + \beta(|\psi\rangle \otimes |\phi_2\rangle). \end{aligned}$$

The resulting vector space is the *tensor product* $\mathcal{H} \otimes \mathcal{K}$. Its elements are sums of product vectors; they need not themselves be a single product vector. We equip it with the inner product determined by

$$\langle \psi \otimes \phi, \psi' \otimes \phi' \rangle = \langle \psi | \psi' \rangle \langle \phi | \phi' \rangle,$$

extended conjugate-linearly in the first argument and linearly in the second. The bilinearity relations ensure that this is well-defined.

Product bases. Choose orthonormal bases $\{|a\rangle\}$ of $\mathcal{H}$ and $\{|b\rangle\}$ of $\mathcal{K}$, and abbreviate $|a\rangle \otimes |b\rangle$ as $|ab\rangle$. Bilinearity gives

$$\left(\sum_a \psi_a |a\rangle \right) \otimes \left(\sum_b \phi_b |b\rangle \right) = \sum_{a,b} \psi_a \phi_b |ab\rangle.$$

Thus the vectors $|ab\rangle$ span $\mathcal{H} \otimes \mathcal{K}$. They satisfy $\langle ab | a'b' \rangle = \delta_{aa'} \delta_{bb'}$, so they form an orthonormal basis. A general vector has the form $|\Psi\rangle = \sum_{a,b} c_{ab} |ab\rangle$; for a single product vector, its coefficients factor as $c_{ab} = \psi_a \phi_b$.

For more factors, we repeat this construction and identify different bracketings by $(u \otimes v) \otimes w \leftrightarrow u \otimes (v \otimes w)$. If $\{|x_j\rangle : x_j \in \Sigma_j\}$ is an orthonormal basis of $\mathcal{H}_j$, then $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$ has the orthonormal basis

$$\{|x_1\rangle \otimes \cdots \otimes |x_n\rangle : x_j \in \Sigma_j\}.$$

We also write $|x_1\rangle \cdots |x_n\rangle$ or $|x_1 \cdots x_n\rangle$. Counting the basis vectors gives

$$\dim(\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n) = \dim(\mathcal{H}_1) \cdots \dim(\mathcal{H}_n).$$

Axiom 4 (Composite systems). If quantum systems have Hilbert spaces $\mathcal{H}_1, \dots, \mathcal{H}_n$, then their joint system has Hilbert space $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$.

Example 2.1. For two qubits the joint Hilbert space is $\mathbb{C}^2 \otimes \mathbb{C}^2$. It is four-dimensional, with standard basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, in this order. For n qubits the Hilbert space is $(\mathbb{C}^2)^{\otimes n}$, with dimension 2^n and standard basis $\{|x_1 \cdots x_n\rangle : x_1, \dots, x_n \in \{0, 1\}\}$.

Notation. We label systems by $A, B, C, \dots$ and their Hilbert spaces by $\mathcal{H}_A, \mathcal{H}_B, \mathcal{H}_C, \dots$. We write $|A| = \dim \mathcal{H}_A$, $\text{Lin}(A) = \text{Lin}(\mathcal{H}_A)$, $\mathcal{S}(A) = \mathcal{S}(\mathcal{H}_A)$, and $\text{PSD}(A) = \text{PSD}(\mathcal{H}_A)$. The label AB denotes the composite system with Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$. Thus $\rho_{AB} \in \mathcal{S}(AB)$ may describe a state shared by Alice and Bob, while μ_A is a measurement on Alice's system. Tensor factors are always written in the order of the subsystem labels.

Tensor products of operators. For $M_A \in \text{Lin}(A)$ and $N_B \in \text{Lin}(B)$, the operator $M_A \otimes N_B \in \text{Lin}(AB)$ is determined by

$$(M_A \otimes N_B)(|\psi_A\rangle \otimes |\phi_B\rangle) = (M_A|\psi_A\rangle) \otimes (N_B|\phi_B\rangle)$$

and linearity. The same definition applies to maps between different input and output spaces. In product bases its entries are

$$(M_A \otimes N_B)_{ab, a'b'} = (M_A)_{a, a'} (N_B)_{b, b'}.$$

This is the Kronecker product of matrices: each entry of M_A is replaced by that entry times the matrix N_B .

Example 2.2. If $M_A = |a\rangle\langle a'|$ and $N_B = |b\rangle\langle b'|$, then

$$M_A \otimes N_B = |a\rangle\langle a'| \otimes |b\rangle\langle b'| = |ab\rangle\langle a'b'|.$$

Definition 2.3. A state of the form $\rho_{AB} = \rho_A \otimes \rho_B$, where $\rho_A \in \mathcal{S}(A)$ and $\rho_B \in \mathcal{S}(B)$, is a *product state*. A state that is not a product state is called *correlated*.

This is a density matrix: the tensor product of positive operators is positive, as follows by taking their spectral decompositions, and

$$\text{tr}[M_A \otimes N_B] = \text{tr}[M_A] \text{tr}[N_B].$$

In particular, $\text{tr}[\rho_A \otimes \rho_B] = 1$.

Example 2.4 (Classical states). Let p_{XY} be a joint probability distribution on $\Sigma_X \times \Sigma_Y$. Its density matrix is

$$\rho_{XY} = \sum_{x,y} p_{XY}(x,y) |xy\rangle\langle xy| = \sum_{x,y} p_{XY}(x,y) |x\rangle\langle x| \otimes |y\rangle\langle y|.$$

Conversely, every density matrix diagonal in this product basis has this form. The state ρ_{XY} is a product state if and only if X and Y are independent under p_{XY} . Product states therefore generalize independence in classical probability.

Example 2.5 (Perfect classical correlation). The classical state

$$\sigma_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|) = \frac{1}{2}(|0\rangle\langle 0| \otimes |0\rangle\langle 0| + |1\rangle\langle 1| \otimes |1\rangle\langle 1|)$$

describes two equal, uniformly random bits. It is not a product state: each bit is uniform, but the outcomes 01 and 10 have probability zero. In the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\sigma_{AB} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Exercise 2.6. Let

$$\rho_A = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \rho_B = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Express $\rho_A \otimes \rho_B$ in the standard basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Solution. Using the block form of the Kronecker product, we obtain

$$\rho_A \otimes \rho_B = \frac{1}{2} \begin{pmatrix} \rho_B & \rho_B \\ \rho_B & \rho_B \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

□

Example 2.7 (Pure product states). For two qubits, the basis vectors $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ are pure product states. Another example is

$$|+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

We freely abbreviate $|\psi_A\rangle \otimes |\phi_B\rangle$ as $|\psi_A\rangle|\phi_B\rangle$.

Example 2.8 (A maximally entangled state). The Bell state

$$|\Phi_{AB}^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

is a pure state that is not a product state. We will explain the term *maximally entangled* in

Section 2.5. Its density matrix is

$$\begin{aligned}\rho_{AB} &= |\Phi_{AB}^+\rangle\langle\Phi_{AB}^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|) \\ &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.\end{aligned}$$

For example, $|00\rangle\langle 11| = |0\rangle\langle 1| \otimes |0\rangle\langle 1|$. The off-diagonal entries distinguish this state from σ_{AB} in Example 2.5.

2.2 The Partial Trace

Classical correlations do not communicate. Suppose a fair coin is tossed and its result is written on two cards, one in Alice's sealed envelope and one in Bob's. Neither has seen the result. Write X for Alice's bit and Y for Bob's. This is the preparation described by the density matrix in Example 2.5. Its four diagonal entries are the *joint* probabilities for 00, 01, 10, 11:

$$(p_{XY}(0,0), p_{XY}(0,1), p_{XY}(1,0), p_{XY}(1,1)) = \left(\frac{1}{2}, 0, 0, \frac{1}{2}\right).$$

Each individual bit is uniform, because we sum over the other bit:

$$p_X(0) = p_{XY}(0,0) + p_{XY}(0,1) = \frac{1}{2} + 0 = \frac{1}{2},$$

and similarly $p_X(1) = p_Y(0) = p_Y(1) = 1/2$. Bob travels far away and opens his envelope. If he sees $Y = 0$, he immediately knows that Alice's card also says 0. This is *conditioning*: $p_{X|Y}(0 | 0) = 1$. He has learned an outcome and updated his prediction about Alice's card. Alice, with her envelope still closed and without Bob's result, still assigns probability $1/2$ to each value of X . Her *marginal distribution* $p_X(x) = \sum_y p_{XY}(x,y)$ is unchanged whether Bob opens his envelope or leaves it closed. She cannot tell which he did, and he cannot choose the random bit he reveals. The correlation was established when the cards were prepared; opening an envelope sends no message.

The quantum no-communication statement below has the same distinction between conditional and marginal probabilities, and holds for every shared state and every choice of local measurement. Its proof makes no assumption that measurement outcomes were fixed in advance.

Local measurements. If Alice performs a measurement $\mu_A : \Omega_A \rightarrow \text{PSD}(A)$, its description on the joint system is the POVM $x \mapsto \mu_A(x) \otimes I_B$. Indeed, its operators are positive and

$$\sum_{x \in \Omega_A} \mu_A(x) \otimes I_B = \left(\sum_{x \in \Omega_A} \mu_A(x) \right) \otimes I_B = I_A \otimes I_B.$$

If Bob also performs a measurement $\mu_B : \Omega_B \rightarrow \text{PSD}(B)$, their joint measurement has effects $\mu_A(x) \otimes \mu_B(y)$ and probabilities

$$p(x,y) = \text{tr}[(\mu_A(x) \otimes \mu_B(y))\rho_{AB}].$$

No-communication for local measurements. For fixed local measurements, $p(x, y)$ is an ordinary joint probability distribution. Alice's outcome distribution is its *marginal*, obtained by summing over Bob's outcome:

$$\begin{aligned} p_A(x) &= \sum_y p(x, y) = \sum_y \text{tr}[(\mu_A(x) \otimes \mu_B(y))\rho_{AB}] \\ &= \text{tr} \left[\left(\mu_A(x) \otimes \sum_y \mu_B(y) \right) \rho_{AB} \right] = \text{tr}[(\mu_A(x) \otimes I_B)\rho_{AB}]. \end{aligned}$$

Thus Alice's outcome distribution does not depend on Bob's choice of measurement: every POVM sums to I_B , so the same marginal results for every choice. The same holds with Alice and Bob interchanged. They cannot send a message merely by choosing a local measurement. Conditioning on an outcome known to Bob can change Alice's conditional statistics, but Alice needs to learn that outcome to use them. This argument concerns outcome probabilities; it requires no rule for the post-measurement state.

At the level of states, the quantum analogue of taking a marginal distribution is the *partial trace*. What state describes Alice's system alone? We seek ρ_A such that

$$\text{tr}[\mu_A(x)\rho_A] = \text{tr}[(\mu_A(x) \otimes I_B)\rho_{AB}]$$

for every local measurement. Choose orthonormal bases $\{|a\rangle\}$ and $\{|b\rangle\}$ of $\mathcal{H}_A$ and $\mathcal{H}_B$. Since $|ab\rangle = (I_A \otimes |b\rangle)|a\rangle$ and $\langle ab| = \langle a|(I_A \otimes \langle b|)$, expanding the trace yields

$$\begin{aligned} \text{tr}[(\mu_A(x) \otimes I_B)\rho_{AB}] &= \sum_{a,b} \langle ab|(\mu_A(x) \otimes I_B)\rho_{AB}|ab\rangle \\ &= \sum_{a,b} \langle a|(I_A \otimes \langle b|)(\mu_A(x) \otimes I_B)\rho_{AB}(I_A \otimes |b\rangle)|a\rangle \\ &= \sum_{a,b} \langle a|\mu_A(x)(I_A \otimes \langle b|)\rho_{AB}(I_A \otimes |b\rangle)|a\rangle \\ &= \sum_a \langle a|\mu_A(x) \left(\sum_b (I_A \otimes \langle b|)\rho_{AB}(I_A \otimes |b\rangle) \right) |a\rangle. \end{aligned}$$

The third line uses $(I_A \otimes \langle b|)(\mu_A(x) \otimes I_B) = \mu_A(x)(I_A \otimes \langle b|)$. The last line is $\text{tr}[\mu_A(x)\rho_A]$ if we set

$$\rho_A = \sum_b (I_A \otimes \langle b|)\rho_{AB}(I_A \otimes |b\rangle).$$

Two qubits in matrix notation. We can obtain the same result using ordinary matrix multiplication. In the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, write

$$\rho_{AB} = \left(\begin{array}{cc|cc} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ \hline r_{31} & r_{32} & r_{33} & r_{34} \\ r_{41} & r_{42} & r_{43} & r_{44} \end{array} \right).$$

For Alice's measurement effect $E = \mu_A(x)$, we have

$$E = \begin{pmatrix} e_{11} & e_{12} \\ e_{21} & e_{22} \end{pmatrix}, \quad E \otimes I_2 = \begin{pmatrix} e_{11} & 0 & e_{12} & 0 \\ 0 & e_{11} & 0 & e_{12} \\ e_{21} & 0 & e_{22} & 0 \\ 0 & e_{21} & 0 & e_{22} \end{pmatrix}.$$

Multiplying the matrices and summing the diagonal entries gives

$$\begin{aligned} \text{tr}[(E \otimes I_2)\rho_{AB}] &= e_{11}(r_{11} + r_{22}) + e_{12}(r_{31} + r_{42}) \\ &\quad + e_{21}(r_{13} + r_{24}) + e_{22}(r_{33} + r_{44}). \end{aligned}$$

This is exactly $\text{tr}[E\rho_A]$ for the 2×2 matrix

$$\rho_A = \begin{pmatrix} r_{11} + r_{22} & r_{13} + r_{24} \\ r_{31} + r_{42} & r_{33} + r_{44} \end{pmatrix}.$$

Thus each entry of ρ_A is the trace of the corresponding 2×2 block of ρ_{AB} . For example, its upper-right entry is

$$r_{13} + r_{24} = \langle 00|\rho_{AB}|10\rangle + \langle 01|\rho_{AB}|11\rangle = \sum_{b=0}^1 \langle 0b|\rho_{AB}|1b\rangle.$$

This is precisely the sum over Bob's basis index in the bra-ket calculation. On the diagonal it gives the usual marginal probabilities for computational-basis measurements; the same summation also determines the off-diagonal entries.

Both calculations motivate the following definition.

Definition 2.9 (Partial trace). For $M_{AB} \in \text{Lin}(AB)$, its *partial trace over B* is

$$\text{tr}_B[M_{AB}] = \sum_b (I_A \otimes \langle b|) M_{AB} (I_A \otimes |b\rangle),$$

where $\{|b\rangle\}$ is any orthonormal basis of $\mathcal{H}_B$. For a state ρ_{AB} , the operator $\rho_A = \text{tr}_B[\rho_{AB}]$ is called its *reduced state* on A .

We will verify shortly that this definition is independent of the basis and that the reduced state is a density matrix. Inserting the definition between $\langle a|$ and $|a'\rangle$ gives its matrix entries:

$$\begin{aligned} \langle a| \text{tr}_B[M_{AB}] |a'\rangle &= \sum_b \langle a| (I_A \otimes \langle b|) M_{AB} (I_A \otimes |b\rangle) |a'\rangle \\ &= \sum_b \langle ab| M_{AB} |a'b\rangle. \end{aligned} \tag{2.1}$$

Equivalently, if

$$M_{AB} = \sum_{a,a',b,b'} M_{ab,a'b'} |a\rangle\langle a'| \otimes |b\rangle\langle b'|,$$

then reconstructing the operator from these matrix entries yields

$$\begin{aligned} \text{tr}_B[M_{AB}] &= \sum_{a,a'} \langle a| \text{tr}_B[M_{AB}] |a'\rangle |a\rangle\langle a'| \\ &= \sum_{a,a'} \left(\sum_b M_{ab,a'b} \right) |a\rangle\langle a'|. \end{aligned}$$

Only entries with the same B index in the row and column contribute. For a tensor product, the same contraction separates into its two factors:

$$\begin{aligned} \text{tr}_B[N_A \otimes O_B] &= \sum_b (I_A \otimes \langle b|) (N_A \otimes O_B) (I_A \otimes |b\rangle) \\ &= \sum_b N_A \langle b| O_B |b\rangle = N_A \text{tr}[O_B]. \end{aligned} \tag{2.2}$$

Similarly, tracing out A gives $\text{tr}_A[N_A \otimes O_B] = \text{tr}[N_A] O_B$. Every operator is a linear combination of tensor products of matrix units, so this rule determines the partial trace on all operators. It also shows that the reduced states of a product state $\rho_A \otimes \rho_B$ are precisely ρ_A and ρ_B , since both factors have trace one.

Lemma 2.10 (Properties of the partial trace).

(a) The map $\text{tr}_B : \text{Lin}(AB) \rightarrow \text{Lin}(A)$ is linear.

(b) For $N_A \in \text{Lin}(A)$ and $M_{AB} \in \text{Lin}(AB)$,

$$\text{tr}[(N_A \otimes I_B)M_{AB}] = \text{tr}[N_A \text{tr}_B[M_{AB}]].$$

(c) The partial trace does not depend on the orthonormal basis of B .

(d) $\text{tr}[\text{tr}_B[M_{AB}]] = \text{tr}[M_{AB}]$.

(e) If $P_{AB} \geq 0$, then $\text{tr}_B[P_{AB}] \geq 0$.

Proof. Part (a) follows from the defining formula. The calculation preceding Definition 2.9 proves (b), with N_A in place of $\mu_A(x)$ and M_{AB} in place of ρ_{AB} .

For (c), suppose that two bases give operators T and T' . By (b), $\text{tr}[N_A(T - T')] = 0$ for every $N_A \in \text{Lin}(A)$. Taking $N_A = |a'\rangle\langle a|$ shows that $\langle a|(T - T')|a'\rangle = 0$ for every pair a, a' , so $T = T'$. Part (d) follows from (b) with $N_A = I_A$.

Finally, for every $|\psi\rangle \in \mathcal{H}_A$,

$$\langle \psi | \text{tr}_B[P_{AB}] | \psi \rangle = \sum_b (\langle \psi | \otimes \langle b |) P_{AB} (| \psi \rangle \otimes | b \rangle) \geq 0,$$

which proves (e). □

Parts (d) and (e) show that a reduced state is a density matrix. Part (b) says that it reproduces every local expectation value: $\mathbb{E}_{\rho_A}[O_A] = \mathbb{E}_{\rho_{AB}}[O_A \otimes I_B]$. It is uniquely determined even by local measurement probabilities, since rank-one projections determine the quadratic form of a Hermitian operator.

Example 2.11 (A reduced state of a Bell pair). For $\rho_{AB} = |\Phi_{AB}^+\rangle\langle\Phi_{AB}^+|$, Eq. (2.2) gives

$$\begin{aligned} \text{tr}_B[\rho_{AB}] &= \frac{1}{2}(|0\rangle\langle 0| \text{tr}[|0\rangle\langle 0|] + |0\rangle\langle 1| \text{tr}[|0\rangle\langle 1|] \\ &\quad + |1\rangle\langle 0| \text{tr}[|1\rangle\langle 0|] + |1\rangle\langle 1| \text{tr}[|1\rangle\langle 1|]) \\ &= \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I_A. \end{aligned}$$

The same calculation gives $\rho_B = I_B/2$. These are also the reduced states of the classically correlated state σ_{AB} and of the product state $(I_A/2) \otimes (I_B/2)$. Alice alone cannot distinguish these three joint preparations, because all her local measurement probabilities are identical.

Mixed states arise from probabilistic preparations and from restricting a joint state to a subsystem, even when the joint state is pure. A measurement also produces a classical outcome distribution, which can be represented by a diagonal density matrix on a classical register. This

last state describes the recorded outcome, not a post-measurement state specified by the POVM alone.

Example 2.12 (Marginal distributions). For the classical joint state

$$\rho_{XY} = \sum_{x,y} p_{XY}(x,y) |x\rangle\langle x| \otimes |y\rangle\langle y|,$$

the reduced state is

$$\rho_X = \text{tr}_Y[\rho_{XY}] = \sum_x \left(\sum_y p_{XY}(x,y) \right) |x\rangle\langle x| = \sum_x p_X(x) |x\rangle\langle x|.$$

Thus partial trace reduces to taking the marginal distribution $p_X(x) = \sum_y p_{XY}(x,y)$. In particular, $p_{XY}(x,y) = p_X(x)p_Y(y)$ for all x,y if and only if $\rho_{XY} = \rho_X \otimes \rho_Y$.

Classical conditional probabilities also make sense after a measurement outcome is specified. For general quantum systems, there is no single measurement-independent conditional density matrix obtained by simply dividing a joint state by a marginal as in classical probability.

Exercise 2.13 (Tensor products of observables). Let $O_A = \sum_x x P_x$ and $O_B = \sum_y y Q_y$ be observables. Show that their local measurements define the joint projective measurement $\{P_x \otimes Q_y\}_{x,y}$ and that

$$\mathbb{E}[xy] = \text{tr}[\rho_{AB}(O_A \otimes O_B)].$$

Show that this expectation factors as $\text{tr}[\rho_A O_A] \text{tr}[\rho_B O_B]$ for a product state. Compute the expectations of $Z \otimes I$, $I \otimes Z$, $Z \otimes Z$, and $X \otimes X$ for σ_{AB} and $|\Phi_{AB}^+\rangle$.

Solution. The operators $P_x \otimes Q_y$ are mutually orthogonal projections summing to I_{AB} . Weighting their outcome probabilities by xy gives

$$\sum_{x,y} xy \text{tr}[\rho_{AB}(P_x \otimes Q_y)] = \text{tr} \left[\rho_{AB} \left(\sum_{x,y} xy P_x \otimes Q_y \right) \right] = \text{tr}[\rho_{AB}(O_A \otimes O_B)].$$

For a product state, multiplication of tensor products and the trace identity give the claimed factorization. In the two specified states, the local Z expectations vanish and the Z outcomes always agree. The operator $X \otimes X$ interchanges $|00\rangle$ and $|11\rangle$, so its diagonal entries on these vectors vanish, while it fixes $|\Phi_{AB}^+\rangle$. Hence

state	$\mathbb{E}[Z \otimes I]$	$\mathbb{E}[I \otimes Z]$	$\mathbb{E}[Z \otimes Z]$	$\mathbb{E}[X \otimes X]$
σ_{AB}	0	0	1	0
$ \Phi_{AB}^+\rangle\langle\Phi_{AB}^+ $	0	0	1	1

□

2.3 Purification

Preparing ρ_x with probability $p(x)$ produces the average state $\rho_A = \sum_x p(x)\rho_x$. As we saw in Chapter 1, the same density matrix can arise from different ensembles.

If we also keep the label x , we describe the preparation using a reference register X , with orthonormal basis $\{|x\rangle : x \in \Omega\}$:

$$\rho_{AX} = \sum_{x \in \Omega} p(x) \rho_x \otimes |x\rangle\langle x|.$$

This is a *classical-quantum state*: X stores classical information about the preparation of A . Its two reduced states are

$$\mathrm{tr}_X[\rho_{AX}] = \rho_A, \quad \mathrm{tr}_A[\rho_{AX}] = \sum_x p(x) |x\rangle\langle x|.$$

Reading X in this basis returns x with probability $p(x)$. Conditional on an outcome with $p(x) > 0$, the state of A is ρ_x . For example, a subsequent POVM μ_A has conditional probabilities $p(y | x) = \mathrm{tr}[\mu_A(y) \rho_x]$. Discarding X loses this side information. We can also use a quantum reference system.

Definition 2.14. A *purification* of $\rho_A \in \mathcal{S}(A)$ is a unit vector $|\phi_{AR}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_R$ such that

$$\mathrm{tr}_R[|\phi_{AR}\rangle\langle\phi_{AR}|] = \rho_A.$$

We also call its density matrix a purification, and R the *reference* or *purifying system*.

Lemma 2.15. Every state ρ_A has a purification. The purifying system can be chosen to have dimension $\mathrm{rank}(\rho_A)$.

Proof. Let $r = \mathrm{rank}(\rho_A)$ and take a spectral decomposition on its support,

$$\rho_A = \sum_{j=0}^{r-1} p_j |e_j\rangle\langle e_j|, \quad p_j > 0.$$

Set $\mathcal{H}_R = \mathbb{C}^r$ and $|\phi_{AR}\rangle = \sum_{j=0}^{r-1} \sqrt{p_j} |e_j\rangle \otimes |j\rangle$. This vector is normalized because $\sum_j p_j = 1$. Moreover,

$$\begin{aligned} \mathrm{tr}_R[|\phi_{AR}\rangle\langle\phi_{AR}|] &= \sum_{j,k=0}^{r-1} \sqrt{p_j p_k} |e_j\rangle\langle e_k| \mathrm{tr}[|j\rangle\langle k|] \\ &= \sum_{j=0}^{r-1} p_j |e_j\rangle\langle e_j| = \rho_A. \end{aligned}$$

□

The calculation only needs the reference vectors $|j\rangle$ to be orthonormal. Any pure-state ensemble $\rho_A = \sum_j p_j |\psi_j\rangle\langle\psi_j|$ therefore gives a purification $\sum_j \sqrt{p_j} |\psi_j\rangle \otimes |j\rangle$. This construction need not use the smallest possible reference system.

Exercise 2.16. Find a purification of

$$\rho_A = \frac{1}{3}(|0\rangle\langle 0| + |+\rangle\langle +| + |1\rangle\langle 1|).$$

Solution. Using the three ensemble labels as an orthonormal basis of R , take

$$|\phi_{AR}\rangle = \frac{1}{\sqrt{3}}(|0\rangle|0\rangle + |+\rangle|1\rangle + |1\rangle|2\rangle).$$

Tracing out R removes the cross terms and gives ρ_A . □

Lemma 2.17 (Uniqueness of purification up to an isometry). *If $|\phi_{AR}\rangle$ and $|\psi_{AS}\rangle$ purify the same state ρ_A and $|R| \leq |S|$, then there is an isometry $V : \mathcal{H}_R \rightarrow \mathcal{H}_S$ such that*

$$(I_A \otimes V)|\phi_{AR}\rangle = |\psi_{AS}\rangle.$$

If $|R| = |S|$, this isometry is unitary.

The proof is left as homework.

2.4 Schmidt Decomposition

Recall the *singular value decomposition* (SVD) from linear algebra. For $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$ of rank r , there are orthonormal bases $\{|e_j\rangle\}$ of $\mathcal{K}$ and $\{|g_j\rangle\}$ of $\mathcal{H}$ and positive numbers $s_1 \geq \dots \geq s_r > 0$ such that

$$M = \sum_{j=1}^r s_j |e_j\rangle\langle g_j|.$$

The s_j are the nonzero singular values. The right singular vectors $|g_j\rangle$ are eigenvectors of $M^\dagger M$, and the left singular vectors $|e_j\rangle$ are eigenvectors of $MM^\dagger$, with eigenvalues s_j^2 . To pair them correctly, take $|e_j\rangle = M|g_j\rangle/s_j$ for $s_j > 0$.

We obtain a corresponding decomposition of a bipartite vector by arranging its coefficients into a matrix.

Theorem 2.18 (Schmidt decomposition). *Let $|\psi_{AB}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ be a pure quantum state. There are orthonormal bases $\{|e_j\rangle\}$ of $\mathcal{H}_A$ and $\{|f_j\rangle\}$ of $\mathcal{H}_B$ and positive numbers $s_1 \geq \dots \geq s_r > 0$, where $r \leq \min(|A|, |B|)$, such that*

$$|\psi_{AB}\rangle = \sum_{j=1}^r s_j |e_j\rangle \otimes |f_j\rangle, \quad \sum_{j=1}^r s_j^2 = 1.$$

The s_j are called Schmidt coefficients and r is the Schmidt rank.

Proof. Fix orthonormal bases $\{|a\rangle\}$ and $\{|b\rangle\}$, and let M be the $|A| \times |B|$ matrix with entries $M_{ab} = \langle ab | \psi_{AB} \rangle$. Choose an SVD $M = \sum_{j=1}^r s_j |e_j\rangle\langle g_j|$ and define

$$|f_j\rangle = \sum_b \overline{\langle b | g_j \rangle} |b\rangle.$$

Thus $|f_j\rangle$ is the coordinatewise conjugate of $|g_j\rangle$ in the chosen basis of B . Conjugation preserves orthonormality, and

$$M_{ab} = \sum_j s_j \langle a | e_j \rangle \langle g_j | b \rangle = \sum_j s_j \langle a | e_j \rangle \langle b | f_j \rangle.$$

Substituting these coefficients into the vector gives

$$\begin{aligned} |\psi_{AB}\rangle &= \sum_{a,b} M_{ab} |a\rangle \otimes |b\rangle \\ &= \sum_j s_j \left(\sum_a \langle a | e_j \rangle |a\rangle \right) \otimes \left(\sum_b \langle b | f_j \rangle |b\rangle \right) = \sum_j s_j |e_j\rangle \otimes |f_j\rangle. \end{aligned}$$

The product vectors in this sum are orthonormal, so normalization implies $\sum_j s_j^2 = 1$. $\square$

Exercise 2.19. Compute the Schmidt decomposition of the state

$$|\psi_{AB}\rangle = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 1 \\ 2 \\ -2 \end{pmatrix}$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Solution. Reshape the vector into its coefficient matrix and compute

$$M = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}, \quad MM^\dagger = \begin{pmatrix} 1/5 & 0 \\ 0 & 4/5 \end{pmatrix}, \quad M^\dagger M = \begin{pmatrix} 1/2 & -3/10 \\ -3/10 & 1/2 \end{pmatrix}.$$

In decreasing order the singular values are $s_1 = 2/\sqrt{5}$ and $s_2 = 1/\sqrt{5}$. The corresponding right singular vectors are $|g_1\rangle = |-\rangle$ and $|g_2\rangle = |+\rangle$, while $M|-\rangle/s_1 = |1\rangle$ and $M|+\rangle/s_2 = |0\rangle$. The right singular vectors are real, so conjugation leaves them unchanged. Hence

$$|\psi_{AB}\rangle = \frac{2}{\sqrt{5}} |1\rangle \otimes |-\rangle + \frac{1}{\sqrt{5}} |0\rangle \otimes |+\rangle.$$

Expanding the two terms recovers the original four entries, including the minus sign on $|11\rangle$. $\square$

Lemma 2.20. If $|\psi_{AB}\rangle = \sum_{j=1}^r s_j |e_j\rangle \otimes |f_j\rangle$ is a Schmidt decomposition, then its reduced states are

$$\rho_A = \sum_{j=1}^r s_j^2 |e_j\rangle \langle e_j|, \quad \rho_B = \sum_{j=1}^r s_j^2 |f_j\rangle \langle f_j|.$$

In particular, both reduced states have rank r and the same nonzero eigenvalues $s_1^2, \dots, s_r^2$, counted with multiplicity.

Proof. Using $\text{tr}[|f_j\rangle \langle f_k|] = \langle f_k | f_j \rangle = \delta_{jk}$ gives

$$\begin{aligned} \rho_A &= \sum_{j,k} s_j s_k |e_j\rangle \langle e_k| \text{tr}[|f_j\rangle \langle f_k|] \\ &= \sum_j s_j^2 |e_j\rangle \langle e_j|. \end{aligned}$$

The same argument with A and B interchanged gives ρ_B . $\square$

Thus the ordered Schmidt coefficients and the Schmidt rank are uniquely determined, though the Schmidt vectors need not be. The lemma also shows that any purification of ρ_A requires $|R| \geq \text{rank}(\rho_A)$. The dimension in Lemma 2.15 is therefore minimal.

Corollary 2.21. *A pure bipartite state is a product state if and only if ρ_A is pure, if and only if ρ_B is pure.*

Proof. A reduced state is pure exactly when its rank, and hence the Schmidt rank, is one. A one-term Schmidt decomposition is a product vector. Conversely, a product vector has pure reduced states. $\square$

2.5 Entanglement

Two classical random variables are independent exactly when their joint state is a product state. Otherwise they are correlated, as in $\sigma_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$. Quantum systems also admit entanglement, a form of correlation that cannot be produced by shared classical randomness and local state preparation. We first define it for pure states.

Definition 2.22. A pure state $|\psi_{AB}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is *entangled* if it is not a product vector: there are no unit vectors $|\psi_A\rangle$ and $|\psi_B\rangle$ with $|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$.

Lemma 2.23. *A pure bipartite state is entangled if and only if either reduced state is mixed, equivalently, if and only if its Schmidt rank is greater than one.*

Proof. This is Corollary 2.21 and Lemma 2.20. $\square$

Definition 2.24. For $|A| = |B| = d \geq 2$, a pure state on AB is *maximally entangled* if both reduced states are maximally mixed:

$$\rho_A = I_A/d, \quad \rho_B = I_B/d.$$

We use the equal-dimension convention here. By Lemma 2.20, a pure state can have both marginals maximally mixed on their whole spaces only when $|A| = |B|$. In equal dimensions it is maximally entangled exactly when it has Schmidt rank d and every Schmidt coefficient is $1/\sqrt{d}$. Thus such states have the form

$$\frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |e_j\rangle \otimes |f_j\rangle,$$

where the two collections are orthonormal bases. In particular, using fixed standard bases, we write

$$|\Phi_{AB}^+\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j\rangle \otimes |j\rangle. \quad (2.3)$$

These states are entangled because their marginals are mixed. Later, entropy will let us quantify the connection between the mixedness of a marginal and the entanglement of a pure joint state.

Lemma 2.25 (Properties of the standard maximally entangled state). *Let $|\Phi_{AB}^+\rangle$ be as in Eq. (2.3). For $d \times d$ matrices M, N expressed in those bases:*

- (a) *The reduced states are $\rho_A = I_A/d$ and $\rho_B = I_B/d$.*
- (b) *$(M \otimes I)|\Phi_{AB}^+\rangle = (I \otimes M^\top)|\Phi_{AB}^+\rangle$.*

$$(c) \langle \Phi_{AB}^+ | M \otimes N | \Phi_{AB}^+ \rangle = \frac{1}{d} \text{tr}[M^\top N] = \frac{1}{d} \text{tr}[MN^\top].$$

Proof. Part (a) follows from Lemma 2.20. For (b),

$$(M \otimes I) | \Phi_{AB}^+ \rangle = \frac{1}{\sqrt{d}} \sum_{a,b} M_{ab} |a\rangle |b\rangle = (I \otimes M^\top) | \Phi_{AB}^+ \rangle.$$

For (c), expand in the same bases:

$$\langle \Phi_{AB}^+ | M \otimes N | \Phi_{AB}^+ \rangle = \frac{1}{d} \sum_{a,b} M_{ab} N_{ab} = \frac{1}{d} \text{tr}[M^\top N] = \frac{1}{d} \text{tr}[MN^\top].$$

□

The transpose in these identities is not the adjoint. For a maximally entangled vector defined using other paired bases $\{|e_j\rangle\}$ and $\{|f_j\rangle\}$, the same identities hold when the matrix entries are taken in those respective bases.

Exercise 2.26. Show that

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle).$$

Solution. The Hadamard matrix $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ satisfies $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, and $HH^\top = I$. By Lemma 2.25(b),

$$(H \otimes H) | \Phi_{AB}^+ \rangle = (I \otimes HH^\top) | \Phi_{AB}^+ \rangle = | \Phi_{AB}^+ \rangle.$$

Applying $H \otimes H$ to the standard-basis expansion proves the claim. For a general unitary U , the corresponding invariance is $(U \otimes \bar{U}) | \Phi_{AB}^+ \rangle = | \Phi_{AB}^+ \rangle$, since $\bar{U}U^\top = I$; one cannot replace $\bar{U}$ by U in general. □

Every classical joint state $\rho_{XY} = \sum_{x,y} p_{XY}(x,y) |x\rangle\langle x| \otimes |y\rangle\langle y|$ is a convex combination of product states. This motivates the definition for arbitrary, possibly mixed, states.

Definition 2.27. A state $\rho_{AB} \in \mathcal{S}(AB)$ is *separable* if there are a finite set Ω , a probability distribution p on Ω , and states $\rho_{A,x} \in \mathcal{S}(A)$ and $\rho_{B,x} \in \mathcal{S}(B)$ such that

$$\rho_{AB} = \sum_{x \in \Omega} p(x) \rho_{A,x} \otimes \rho_{B,x}.$$

A state is *entangled* if it is not separable.

Operationally, Alice and Bob can prepare a separable state by sharing a classical random variable x , then locally preparing $\rho_{A,x}$ and $\rho_{B,x}$. Product states are separable, but separable states can be correlated: σ_{AB} is an example. States diagonal in a product basis are also separable. The converse does not hold in general, since the local states in a separable decomposition need not be diagonal in common local bases. In particular, an entangled state cannot be diagonal in any product basis.

The two definitions of entanglement agree on pure states. To see this, suppose $|\psi\rangle\langle\psi| = \sum_x p(x)\tau_x$ with $\tau_x = \rho_{A,x} \otimes \rho_{B,x}$ and $p(x) > 0$. For every $|v\rangle$ orthogonal to $|\psi\rangle$,

$$0 = \sum_x p(x) \langle v | \tau_x | v \rangle.$$

Each summand is nonnegative, so $\langle v | \tau_x | v \rangle = \|\sqrt{\tau_x}|v\rangle\|^2 = 0$ for every x . Thus every τ_x is supported on the one-dimensional span of $|\psi\rangle$ and, having trace one, equals $|\psi\rangle\langle\psi|$. Consequently this pure state is a product state. Its factors are pure, because

$$1 = \text{rank}(\rho_{A,x} \otimes \rho_{B,x}) = \text{rank}(\rho_{A,x}) \text{rank}(\rho_{B,x}).$$

Conversely, any pure product state is separable by definition.

Overview of bipartite states. Figure 2.1 summarizes these distinctions. Here, as for classical registers earlier in the notes, *classical* means diagonal in a *fixed* product basis:

$$\rho_{AB} = \sum_{a,b} p(a,b) |a\rangle\langle a| \otimes |b\rangle\langle b|.$$

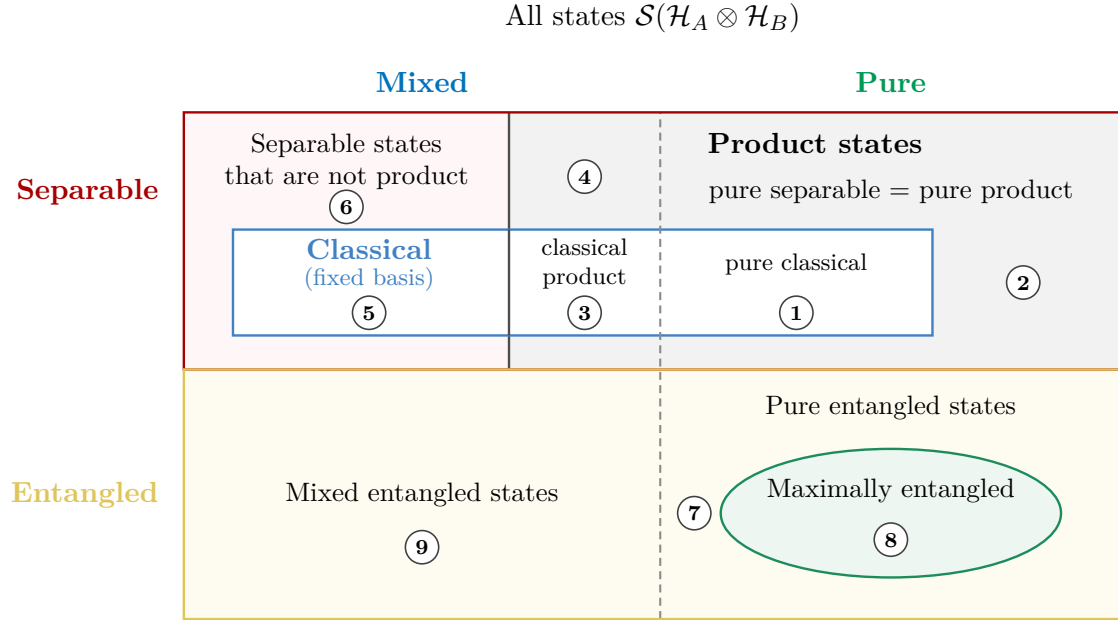
Every such classical state is separable. It is a product state exactly when $p(a,b) = p_A(a)p_B(b)$, and it is pure exactly when one probability equals one. Thus pure classical states are the projectors onto the chosen product-basis vectors, whereas pure separable states are all pure product states. Product states need not be diagonal in this fixed basis: for example, $|+\rangle\langle+| \otimes I/2$ is not diagonal in the computational basis.

If instead we allow the local bases to depend on the state, every product state is diagonal in some product basis: take eigenbases of its two factors. Under that convention, the product states form a subset of the classical states. The fixed-basis convention in the figure is therefore essential.

For entry 6 in Fig. 2.1, separability follows from the displayed mixture of product states. The vectors $|00\rangle$ and $|1+\rangle$ are orthogonal, so τ_{AB} has rank two and is mixed. Its marginals are

$$\tau_A = I_2/2, \quad \tau_B = \frac{1}{2}(P_0 + P_+) = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}.$$

Both marginals have rank two, so their product has rank four and cannot equal τ_{AB} . Thus the state is correlated. It is nonclassical in the fixed computational basis because $\langle 10 | \tau_{AB} | 11 \rangle = 1/4$ is nonzero. The following exercise justifies entry 9 and illustrates why reduced states alone do not determine whether a mixed joint state is entangled.



Write $P_v = |v\rangle\langle v|$ and $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

No.	State	Classification
1	P_{00}	Pure, classical, product
2	P_{+0}	Pure, nonclassical, product
3	$I_4/4$	Mixed, classical, product
4	$P_+ \otimes I_2/2$	Mixed, nonclassical, product
5	$\sigma_{AB} = \frac{1}{2}(P_{00} + P_{11})$	Mixed, classical, correlated
6	$\tau_{AB} = \frac{1}{2}(P_{00} + P_{1+})$	Mixed, nonclassical, separable, correlated
7	$P_\psi, \psi\rangle = \frac{\sqrt{3}}{2} 00\rangle + \frac{1}{2} 11\rangle$	Pure entangled, not maximally entangled
8	P_{Φ^+}	Maximally entangled
9	$\frac{1}{2}\sigma_{AB} + \frac{1}{2}P_{\Phi^+}$	Mixed entangled

Figure 2.1: Schematic relationships between two-qubit state classes. The numbered points identify the examples in the table. Classical means diagonal in the fixed computational product basis. Every pure separable state is a product state. Shapes and areas do not describe the geometry or relative sizes of the sets.

Exercise 2.28 (Same reduced states, different correlations). Let $\sigma_{AB} = \frac{1}{2}(P_{00} + P_{11})$, where $P_v = |v\rangle\langle v|$, and let $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. For $0 \leq t \leq 1$, consider

$$\rho_t = (1-t)\sigma_{AB} + tP_{\Phi^+} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & t \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ t & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find the eigenvalues of ρ_t . For which t is it pure? For which t is it classical in the computational product basis?
- (b) Compute both reduced states. Is ρ_t ever a product state?
- (c) Prove that every separable two-qubit state ω satisfies

$$\text{tr}[\omega(X \otimes X + Z \otimes Z)] \leq 1.$$

Hint: First use Bloch vectors and Cauchy–Schwarz for product states, then use convexity.

- (d) Evaluate this expression for ρ_t , determine exactly when it is entangled, and locate the family in Fig. 2.1.

Solution. (a) The nonzero block on the span of $|00\rangle, |11\rangle$ has eigenvectors $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$. The four eigenvalues are

$$\frac{1+t}{2}, \quad \frac{1-t}{2}, \quad 0, \quad 0.$$

Thus ρ_t is pure exactly when $t = 1$; equivalently, $\text{tr}[\rho_t^2] = (1+t^2)/2$ equals one exactly then. Its only possible off-diagonal entries are $t/2$, so it is classical in the fixed basis exactly when $t = 0$.

(b) Tracing out either qubit removes the off-diagonal terms, leaving

$$(\rho_t)_A = (\rho_t)_B = I_2/2 \quad \text{for every } t \in [0, 1].$$

If ρ_t were a product state, it would equal the product of its marginals, $I_4/4$. This is impossible: its probability of 01 in the computational basis is zero, whereas that probability is $1/4$ for $I_4/4$.

(c) For a product state $\alpha \otimes \beta$, write the two Bloch vectors as $a = (a_x, a_y, a_z)$ and $b = (b_x, b_y, b_z)$, with $\|a\|, \|b\| \leq 1$. By the factorization of product-state expectations,

$$\begin{aligned} \text{tr}[(\alpha \otimes \beta)(X \otimes X + Z \otimes Z)] &= a_x b_x + a_z b_z \\ &\leq \sqrt{a_x^2 + a_z^2} \sqrt{b_x^2 + b_z^2} \leq 1. \end{aligned}$$

If $\omega = \sum_j p_j \alpha_j \otimes \beta_j$ is separable, linearity of the trace gives an average of quantities at most one. This proves the same bound for ω .

(d) The state ρ_t is supported on $|00\rangle, |11\rangle$, both of which have $Z \otimes Z$ eigenvalue $+1$. The operator $X \otimes X$ interchanges these two vectors, so

$$\text{tr}[\rho_t(Z \otimes Z)] = 1, \quad \text{tr}[\rho_t(X \otimes X)] = t.$$

Their sum is $1 + t$, which exceeds the separable bound whenever $t > 0$. For $t = 0$, the state is the displayed separable mixture σ_{AB} . Therefore the complete classification is

$$\begin{array}{l|l} t = 0 & \text{mixed, classical, separable, correlated (point 5),} \\ 0 < t < 1 & \text{mixed entangled (point 9 represents } t = 1/2\text{),} \\ t = 1 & \text{pure maximally entangled (point 8).} \end{array}$$

The reduced states are the same throughout, although the joint correlations change.

The observable $W = I_{AB} - X \otimes X - Z \otimes Z$ has nonnegative expectation on every separable state and expectation $-t$ on ρ_t . Such an observable is called an *entanglement witness*: a negative expectation certifies entanglement.¹ $\square$

¹This witness is the two-qubit case of Eq. (4) in G. Tóth and O. Gühne, *Two measurement settings can suffice to verify multipartite entanglement*, [arXiv:quant-ph/0409132](https://arxiv.org/abs/quant-ph/0409132). The proof here uses only Bloch vectors and convexity.